Data Science Capstone — Literature Review

Project: MoodBand: An Inclusive Wearable System for Real-Time Mood Detection and Adaptive Sensory Feedback

Supervisor: Prof. Rached Zantout

Student: Ritik Pravin Mota

1. Affective Computing Fundamentals

Picard's foundational work [1] is where affective computing as a discipline really starts. Her central claim was that computers could be built to recognize and respond to emotional states, not just ignore them as noise. That idea sounds almost obvious now, but at the time it wasn't, and it's still the premise MoodBand rests on: that internal affective states leave measurable traces a system can learn to read.

Decades later, Conceição et al. [2] show what that premise looks like in practice with modern tools. They build on Russell's circumplex model, which maps emotion along two axes, valence and arousal, and use a deep-seeded clustering algorithm to classify emotional states from signals collected by a wearable wristband, testing it across three separate benchmark datasets. What makes this paper useful for MoodBand specifically is that it's working with the same kind of hardware form factor (a wrist-worn device) and the same underlying signal types we'd expect from a Fitbit, and it still manages to map those signals onto discrete emotional categories reasonably well.

Implication for MoodBand: together, [1] and [2] justify the project's core premise — that physiological signals can be mapped onto discrete emotional categories — and specifically justify grounding MoodBand's mood labels (happy, sad, stressed, angry, neutral) in the valence/arousal circumplex model rather than inventing an ad hoc labelling scheme. This directly shaped the mood categories used throughout the system design.

2. Wearable Physiological Sensing

Moving from theory to the specific question of wearables, Saganowski et al.'s systematic review [3] is a useful starting point. They find that heart rate, HRV, and electrodermal activity dominate the literature as the go-to signals for everyday emotion recognition, and they flag how much harder things get once you move out of the lab and into real, everyday conditions. That's exactly the setting MoodBand is targeting.

On the consumer-hardware side specifically, [4] is probably the single most reassuring paper on this list. Using the LifeSnaps dataset, built from actual Fitbit Sense smartwatches rather than research-grade sensors, the authors predict anxiety from everyday wearable data and get results suggesting a plain consumer device carries useful signal, not just noise.

That said, [5] is a good reality check on the same point. The authors ran into some genuinely annoying constraints when pulling Fitbit data for ML purposes: step counts reading zero while the device is just charging, HR data missing when the device isn't worn properly, all requiring manual filtering. Most notably, Fitbit only computes HRV while the user is asleep and relatively still, which would otherwise mean HRV could not be relied upon as a live, daytime signal.

Implication for MoodBand: papers [4] and [5] both use Fitbit devices specifically, which is directly relevant since Fitbit is also the device selected for this project. The choice of Fitbit was driven primarily by a requirement outside the scope of this literature review on its own: the companion app must work for both Android and iOS users. Fitbit's cloud-based API serves data identically regardless of the user's phone platform, whereas device-native alternatives such as Apple's HealthKit only work on iOS. This means the HRV constraint identified in [5] — that Fitbit only computes HRV overnight — carries over directly into MoodBand's design, and the classification pipeline is built around that constraint rather than around it being designed away.

3. Mood/Emotion Classification Models

Once the signals are available, the question is what model actually does the classifying. The systematic review in [6] gives a pretty clear answer: classical machine learning methods, Random Forest and Support Vector Machines in particular, show up in the large majority of wearable-based stress detection studies, with Random Forest especially holding up well on EDA and HRV features.

[7] adds two things on top of that. First, the authors design their classification approach specifically for resource-constrained, on-device deployment — directly relevant given MoodBand's classifier must run on a mobile phone rather than a server. Second, they compare personalized models against generalized ones and find personalized models can win out even with limited data per person, which matters given MoodBand's own user trial will necessarily be small.

Implication for MoodBand: [6] directly influenced the choice of classifier family — Random Forest is used as the primary candidate classifier specifically because it is the best-performing classical model on EDA/HRV-style wearable features in the reviewed literature, rather than defaulting to a heavier deep-learning approach that would be harder to justify running on a phone. [7] influenced two further decisions: favouring lightweight, resource-constrained model architectures suitable for on-device or near-device inference, and adopting a personalized-model evaluation strategy alongside a generalized one, given the project's necessarily small participant sample.

4. Accessible Feedback Design

The last piece of MoodBand, getting feedback to the user through non-visual channels, is backed by a smaller but still useful set of papers. The iFeeling system [8] is about as close a precedent as exists: it takes a detected emotional state and renders it as a distinct vibration pattern on a phone, built explicitly with visually impaired users in mind.

[9] widens the lens a bit. It's a recent, large-scale systematic literature review spanning two decades (2004 to 2024) of haptic assistive tools for blind and low-vision users, drawing on 132 papers, and it's useful less for any single finding and more as a way to check iFeeling's design choices against the broader literature.

Finally, [10] grounds all of this in actual user experience. Yeo and Chew interviewed visually impaired people in Singapore about how they use mobile apps day to day, and the picture isn't entirely rosy: screen readers often don't work as reliably as one would hope. Their recommendations explicitly call for combining audio and haptic feedback rather than relying on either alone, and for taking users' emotional wellbeing seriously as a design goal.

Implication for MoodBand: [8] directly shaped the haptic feedback channel — MoodBand's vibration-based mood cues are modelled on iFeeling's approach of rendering a detected emotional state as a distinct phone-vibration pattern. [9] broadened this into a fuller design vocabulary: MoodBand's use of graduated or directional vibration patterns (e.g., descending vibration intensity for anger) draws on patterns documented across this wider body of research rather than the single iFeeling precedent alone. [10] most directly shaped a specific design rule: because screen readers were found to behave unreliably in practice, MoodBand pairs every visual cue with a corresponding audio or haptic equivalent by default, rather than treating a screen-reader-compatible UI as sufficient on its own.

5. Mood-Regulation Interventions and Feedback Delivery

The sections above cover how to detect a mood and how to deliver feedback non-visually, but not which intervention actually helps with a given mood, or when it's appropriate to say anything at all. That gap matters more than it might first appear: MoodBand's current mood classifier, a Random Forest trained on heart rate and step data from the 30 minutes before each self-reported mood in the LifeSnaps dataset, reaches about 0.33 balanced accuracy against a 0.25 chance baseline across four classes. That's real signal, but modest enough that acting on every prediction as if it were certain would be a mistake. This section looks at what the literature says about choosing and timing interventions, with that constraint specifically in mind.

Nahum-Shani et al. [11] provide the organizing framework. They define a just-in-time adaptive intervention (JITAI) by six components: a distal outcome, proximal outcomes, decision points, intervention options, tailoring variables, and decision rules — with tailoring variables capturing either vulnerability/opportunity (is intervention needed right now) or receptivity (can the person actually engage with it right now). It's a structure built for exactly this kind of

problem: a system that has to decide, repeatedly, whether and how to intervene based on an imperfect read of someone's state.

Implication for MoodBand: [11] directly shapes how the feedback-decision module is framed — not as a simple mood-to-feedback lookup table, but as a JITAI decision rule, where each new prediction is a decision point, mood and context (e.g., recent activity) are the tailoring variables, and the feedback catalog is the set of intervention options.

Several papers speak directly to what goes wrong when that decision rule is too eager. Koh et al.'s systematic review [18] screened 2,152 studies down to 52 on stress-tracking and intervention systems, and found just-in-time delivery to be the most common approach (30 of 52), with on-demand delivery close behind (24). More usefully for MoodBand, it also catalogs what tends to fail: interventions delivered at unwanted moments cause fatigue and attrition, a "perceptual mismatch" between what the system flags and what the user is actually experiencing (their example: exercise flagged as stress) erodes trust, and detection on its own explains what state someone is in but not why. Its main recommendation is human-in-the-loop calibration — letting users correct the system's read of their state. A pilot study on wearable-based self-management during an endurance-cycling program [19] (a preprint, not yet peer-reviewed) backs this up anecdotally: participants generally trusted the detection but some found the alerts "overly sensitive," and preferred brief, easy-to-interpret notifications over detailed ones.

Implication for MoodBand: given MoodBand's own classifier accuracy is modest, [18] and [19] together justify building several guardrails into the feedback-decision module rather than treating every prediction as actionable. Concretely: the system only acts when the model is confident and the mood has persisted across several recent predictions, rather than on a single reading; interventions are rate-limited (a default 30-minute cooldown, a daily cap, quiet hours, and a user-controlled snooze); a detected "stressed" state is suppressed if recent step data shows the user has just been physically active, to avoid exactly the perceptual-mismatch failure [18] describes; feedback is framed as a suggestion ("Want a 1-minute breathing break?") rather than a diagnostic claim ("You are stressed"); and every intervention carries a simple "Was this right?" (yes/no/snooze) response, which both respects [18]'s human-in-the-loop recommendation and doubles as in-the-moment labelled data for the personalized-model strategy already motivated by [7]. An on-demand "I need a break" option is offered alongside these automatic triggers, reflecting [18]'s finding that on-demand delivery is nearly as common as just-in-time delivery in practice.

For the stress intervention itself, several lines of evidence point toward a breathing-based response. Two meta-analyses reach somewhat different conclusions on how strong the effect is: Fincham et al. [12], pooling 12 randomized-controlled trials and 785 adults, found breathwork reduced self-reported stress versus controls ($g \approx -0.35$, with secondary reductions in anxiety and depressive symptoms); Shao et al. [13], a larger pool of 31 studies and 1,133 non-clinical participants, found clear cardiovascular effects from slow-paced breathing but only a marginal effect on perceived stress (SMD ≈ -0.51, $p = .06$) — a useful reminder that the physiological and the self-reported effect aren't always the same size. On the delivery side, two small HCI studies suggest a wearable can actually guide this without a screen: Miri et al.'s vibrotactile breathing pacer [14] reduced anxiety under a cognitive stressor relative to a control ($d \approx .33$, $p = .004$), with strong-exhale-phased patterns working best and personalization mattering; and Costa et al.'s EmotionCheck [15] found that wrist vibrations paced slightly slower than a user's actual heart rate reduced anxiety in 67 participants, even though the feedback was false. A field study with law-enforcement officers [20] adds a practical data point: a smartwatch that prompted breathing or meditation when heart rate stayed elevated was rated useful by most of the 22 officers interviewed, particularly the vibration notification itself, though how often people actually opened the notification varied widely. Separately, de Witte et al.'s meta-analysis of 104 RCTs and 9,617 participants [16] found music interventions reduce stress-related outcomes generally (physiological $d \approx .38$, psychological $d \approx .55$), with the single largest physiological effect specifically on heart rate ($d \approx .46$).

Implication for MoodBand: together, [12]–[15] and [20] are why the stressed-mood intervention is a breathing pacer at roughly six breaths per minute, delivered as a strong-exhale-phased vibration pattern by default, with a slow, heartbeat-like tap offered as an alternative — both patterns directly modelled on what [14] and [15] found to work without a screen. [16] is why music is offered as the audio option for stress specifically, rather than an arbitrary playlist, since it showed the largest effect on heart rate of the interventions reviewed here.

Sadness is covered by a single, smaller study. Starcke et al. [17] induced sadness experimentally and then played participants one of several music sequences; a sequence that started sad and gradually became happier (the "iso

principle") left people with higher positive affect and valence and lower negative affect than the comparison sequences, though not every pairwise difference reached significance and the sample was small.

Implication for MoodBand: [17] is the reason the sad-mood audio intervention is an iso-principle sequence — music that matches the user's current low mood first, then gradually brightens — rather than the uplifting playlist originally proposed. Given this rests on one small study, it's treated as a reasonable starting point to test in the user trial, not a settled finding.

Two further findings bear on when to say nothing at all. Koh et al. [18] note that happy or neutral states are usually better left unacknowledged, since repeated positive confirmations risk the same fatigue problem documented for over-frequent warnings. A pair of two-week University of Twente student-thesis studies on wearable stress feedback [21] — a weaker, non-peer-reviewed source, flagged here as such — found that across two cohorts ($n \approx 70$ and $n \approx 26$) stress feedback did not raise average perceived stress, but that some individuals did feel more stressed after receiving it, a cautionary note that reinforces [18]'s suggestion-rather-than-diagnosis recommendation above.

Implication for MoodBand: [18] is why happy and neutral states trigger no feedback at all while they continue, a departure from the proposal's original plan of a green confirmation banner or pleasant chime for these states; [21]'s mixed finding is part of why feedback throughout is framed as an optional suggestion rather than an assertion about the user's state.

Two other planned behaviors have no direct support in the sources reviewed here, and that's worth being explicit about rather than implying otherwise. No source addresses anger-specific interventions, so the calming-timer response kept for anger remains a design choice carried over from the original proposal — to be evaluated in the user trial rather than assumed correct, and relevant only once the classifier itself supports an anger label. Similarly, no source here speaks to whether positive reinforcement should be delivered specifically on recovery from a negative state to a neutral or positive one; restricting feedback to transitions rather than steady states is motivated by the fatigue concern above, not by direct evidence that reinforcement itself helps, and is likewise something to check empirically rather than assume.

Finally, the question of which channel to use for which user is already addressed by the accessible-feedback literature in Section 4, but two of the stress papers reinforce it from a different angle: both Miri et al.'s pacer [14] and Costa et al.'s EmotionCheck [15] regulate affect through vibration alone, with no visual or auditory component at all, which is direct evidence that haptic-only regulation is viable rather than a fallback for when a screen isn't usable.

Implication for MoodBand: combined with [8], [9], and [10] from Section 4, this is why MoodBand defines three accessibility profiles rather than a single sighted/non-sighted split: sighted users get visual feedback with haptic by default; low-vision users get audio and haptic, with high-contrast visuals available optionally; and blind or visually impaired users always get audio and haptic together, never one alone. Every visual cue keeps a non-visual equivalent across all three profiles.

6. Synthesis

Put together, the pieces MoodBand depends on each have reasonable support behind them. Physiological signals can be mapped onto discrete moods [1], [2]. A consumer wearable like Fitbit is a workable, if imperfect, way to collect those signals [3]–[5]. Classical, lightweight models are a sensible fit for the classification step given the deployment constraints [6], [7]. Haptic feedback in particular has real precedent for communicating affective state to visually impaired users [8]–[10]. And a growing body of intervention research supports specific, non-visual responses to at least two of the four target moods — breathing-based feedback for stress and iso-principle music for sadness — along with clear guidance on when not to intervene at all [11]–[21], even though two gaps remain: no evidence here addresses anger-specific interventions or positive reinforcement on recovery, and those pieces are carried as explicit design choices rather than literature-backed ones. What doesn't exist yet, as far as this review found, is a single system that ties all five pieces together into one mobile-first pipeline — a gap this project aims to address.

7. Sensor Recommendations Across Reviewed Literature

Table 2 maps each reviewed paper to the physiological sensors or signals it used or recommended, alongside the sensors MoodBand will actually move forward with.

Paper	Sensors Used / Recommended
[1] Picard (1997)	N/A — conceptual/foundational text, no specific sensor
[2] Conceição et al.	Heart rate / blood-volume pulse (PPG), electrodermal activity (EDA), skin temperature (via Empatica E4 wristband)
[3] Saganowski et al. (review)	Heart rate, HRV, electrodermal activity (identified as the dominant signals across the wearable-sensing literature)
[4] Velu et al. (LifeSnaps/Fitbit)	Heart rate, Fitbit-derived stress score, sleep, step count (Fitbit Sense)
[5] King et al.	Heart rate, HRV (sleep-only), step count / motion (Fitbit intraday data)
[6] Pataca et al. (review)	Electrodermal activity, HRV (found Random Forest performs best specifically on these two)
[7] Irfan et al.	ECG (minimal wearable sensing)
[8] iFeeling	N/A — haptic output/rendering paper, not a sensing paper
[9] Jiang et al. (review)	N/A — haptic feedback review, not sensing
[10] Yeo & Chew	N/A — qualitative UX study, not sensing
Sensors MoodBand will use	**Heart rate (continuous, real-time) and motion/accelerometer (continuous), with heart-rate variability (HRV) incorporated as a supplementary overnight-only signal**

Table 2: Sensors recommended per paper vs. sensors selected for MoodBand

These are the signals consistently available through a consumer-grade, cross-platform-friendly wearable API (Fitbit, via the Google Health API). Electrodermal activity and skin temperature — used in the lab-grade Empatica E4 wristband in [2] and recommended generally in [3] and [6] — are not exposed by any mainstream consumer wearable's public API and are therefore out of reach for this project. ECG-based sensing as explored in [7] likewise requires dedicated ECG-capable hardware (e.g., a chest strap, or Apple Watch's ECG app) not available through Fitbit's consumer API. MoodBand's sensor selection therefore reflects the intersection of what the literature recommends and what a consumer-grade, cross-platform wearable API actually exposes, with HRV demoted to a supplementary overnight signal per the constraint identified in [5].

8. Device Selection Summary

To select a specific wearable, four candidate devices — Fitbit, Apple Watch, Samsung Galaxy Watch, and WHOOP — were compared against price, cross-platform support, developer API accessibility, real-time data availability, and development-environment fit. This section summarizes that comparison and its conclusion.

One piece of context shaped the timing of this decision: the legacy Fitbit Web API (the one referenced by [4] and [5]) was retired by Google — developer support ended September 30, 2026, with full shutdown October 30, 2026 — and replaced by the new Google Health API. This is a net positive for the project: the legacy API required a justification request to unlock sub-daily heart-rate data, whereas the new API provides 5-second-resolution heart rate by default with no special approval step.

Criterion	Fitbit	Apple Watch (SE 3)	Samsung Galaxy Watch	WHOOP
Price (new / used)	$70–130 / $30–80	$220–250 / $120–180	$170–310 / $100–150	Bundled with membership

Criterion	Fitbit	Apple Watch (SE 3)	Samsung Galaxy Watch	WHOOP
Works on Android?	Yes	No — iPhone-only, structurally	Yes	Yes
Works on iOS?	Yes	Yes	No — continuous streaming is Android-only	Yes
API approval needed?	No (new Google Health API)	No (HealthKit)	Only for Samsung's cloud API (1–2 wks)	Free API, but returns no data without paid membership
Real-time daytime HR?	Yes — 5-sec resolution	Yes (workout mode)	Yes — on-device Sensor SDK	No — no continuous/raw HR via API
Matches lit. review?	Yes — [4], [5] used Fitbit directly	No direct precedent	No direct precedent	No direct precedent

Table 3: Condensed device comparison

Decision: Fitbit, integrated via the Google Health API, was finalized as the project's sensing device. The deciding factor was the hard requirement that the companion app support both Android and iOS: this structurally rules out Apple Watch (pairs with iPhone only) and Samsung Galaxy Watch (continuous real-time streaming is Android-only via the Samsung Health Sensor SDK), regardless of their other technical merits. Fitbit's cloud-based API is the only option on the shortlist that serves identical data regardless of phone platform. WHOOP was ruled out independently of the platform question, since its API exposes no continuous or raw heart-rate data under any circumstance and requires an active paid membership just to return data at all.

References

[1] R. W. Picard, Affective Computing. Cambridge, MA, USA: MIT Press, 1997.

[2] M. A. Conceição, A. Dubois, S. Haustein, B. Miranda, and C. L. Azevedo, “Deep-seeded clustering for emotion recognition from wearable physiological sensors,” arXiv preprint arXiv:2308.09013, 2023 (revised 2025).

[3] S. Saganowski, B. Perz, A. G. Polak, and P. Kazienko, “Emotion recognition for everyday life using physiological signals from wearables: A systematic literature review,” IEEE Trans. Affective Comput., vol. 14, no. 3, pp. 1876–1897, 2023.

[4] A. Velu, J. Ramesh, A. Ahmed, S. Ganguly, R. Aburukba, A. Sagahyroon, and F. Aloul, “Integrating Fitbit wearables and self-reported surveys for machine learning-based state–trait anxiety prediction,” Appl. Sci., vol. 15, no. 19, p. 10519, 2025.

[5] Z. D. King, M. Khalid, H. Yu, K. Shibuya, K. Zanna, M. Majd, R. L. Brown, Y. Shen, G. Kypriotakis, T. Vaessen, C. P. Fagundes, and A. Sano, “Machine learning-based context aware EMAs: An offline feasibility study,” arXiv preprint arXiv:2506.15834, 2025.

[6] A. O. Pataca, E. Zdravevski, P. J. Coelho, N. M. Garcia, M. Deryuck, C. Albuquerque, and I. M. Pires, “Use of machine learning for predicting stress episodes based on wearable sensor data: A systematic review,” Comput. Biol. Med., vol. 198, pt. A, p. 111166, 2025.

[7] M. Irfan, A. Nawaz, A. K. Bulbul, R. Klén, A. Subasi, T. Westerlund, and W. Chen, “Emotion recognition with minimal wearable sensing: Multi-domain feature, hybrid feature selection, and personalized vs. generalized ensemble model analysis,” in Proc. 2025 IEEE Int. Conf. Bioinformatics and Biomedicine (BIBM), 2025, pp. 5447–5453.

[8] S. ur Réhman and L. Liu, “iFeeling: Vibrotactile rendering of human emotions on mobile phones.”

[9] C. Jiang, E. Kuang, and M. Fan, “How can haptic feedback assist people with blind and low vision (BLV): A systematic literature review,” arXiv preprint arXiv:2412.19105, 2024.

[10] Y. P. H. Jesvin and C. K. Hao, “A framework for inclusive mobile application design: Insights from Singapore's visually impaired community,” ADADA, vol. 29, suppl. 1, pp. 46–XX, 2024.

[11] I. Nahum-Shani, S. N. Smith, B. J. Spring, L. M. Collins, K. Witkiewitz, A. Tewari, and S. A. Murphy, “Just-in-time adaptive interventions (JITAIs) in mobile health: Key components and design principles for ongoing health behavior support,” Ann. Behav. Med., vol. 52, no. 6, pp. 446–462, 2018, doi: 10.1007/s12160-016-9830-8.

[12] G. W. Fincham, C. Strauss, J. Montero-Marin, and K. Cavanagh, “Effect of breathwork on stress and mental health: A meta-analysis of randomised-controlled trials,” Sci. Rep., vol. 13, p. 432, 2023, doi: 10.1038/s41598-022-27247-y.

[13] R. Shao, I. S. C. Man, and T. M. C. Lee, “The effect of slow-paced breathing on cardiovascular and emotion functions: A meta-analysis and systematic review,” Mindfulness, vol. 15, no. 1, pp. 1–18, 2024, doi: 10.1007/s12671-023-02294-2.

[14] P. Miri, E. Jusuf, H. Margarit, R. Flory, A. Uusberg, K. Isbister, K. Marzullo, and J. J. Gross, “Evaluating a personalizable, inconspicuous vibrotactile (PIV) breathing pacer for in-the-moment affect regulation,” in Proc. CHI Conf. Human Factors Comput. Syst., 2020, doi: 10.1145/3313831.3376757.

[15] J. Costa, A. T. Adams, M. F. Jung, F. Guimbretière, and T. Choudhury, “EmotionCheck: A wearable device to regulate anxiety through false heart rate feedback,” GetMobile, vol. 21, no. 2, pp. 22–25, 2017, doi: 10.1145/3131214.3131222.

[16] M. de Witte, A. Spruit, S. van Hooren, X. Moonen, and G.-J. Stams, “Effects of music interventions on stress-related outcomes: A systematic review and two meta-analyses,” Health Psychol. Rev., vol. 14, no. 2, pp. 294–324, 2020, doi: 10.1080/17437199.2019.1627897.

[17] K. Starcke, J. Mayr, and R. von Georgi, “Emotion modulation through music after sadness induction—The iso principle in a controlled experimental study,” Int. J. Environ. Res. Public Health, vol. 18, no. 23, p. 12486, 2021, doi: 10.3390/ijerph182312486.

[18] Y. Koh, J. Kim, K. Lee, Y. Jung, H. Hong, and U. Lee, “Exploring data-driven approaches to stress management: A systematic review of stress tracking, intervention, and system evaluation methods,” in Proc. CHI Conf. Human Factors Comput. Syst., 2026, doi: 10.1145/3772318.3791194.

[19] “Ride, track, and recover: Pilot randomized trial of a wearable digital self-management intervention during a veteran endurance-cycling program,” arXiv:2606.13529, 2026.

[20] K. K. Jetelina, R. Molsberry, L. Malthaner, A. Beauchamp, M. B. Cannell, T. Hall, E. Fowler, and L. Anderson, “Acceptability of a real-time notification of stress and access to self-help therapies among law enforcement officers,” BMC Public Health, vol. 22, p. 32, 2022, doi: 10.1186/s12889-021-12423-y.

[21] University of Twente student theses on wearable stress feedback, 2024–2025 (e.g., essay.utwente.nl/essays/107074).